

Aaryan Patel

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EDUCATION

University of Wisconsin-Madison

Madison, Wisconsin

B.S. Computer Science and Data Science — Minors in Entrepreneurship and Game Design

Expected May 2027

Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Machine Learning for Engineering Research, Building User Interfaces, Computer Engineering, Data Science Modeling, Linux and C Programming

SKILLS

Languages: Java, Python, R, JavaScript, TypeScript, HTML, CSS, Bash, SQL

Tools: React, AWS (Lambda, S3, DynamoDB, Kendra, Bedrock), Git, Spark, Terraform, Pandas, Spring Boot, Jenkins

Certifications: Oracle Cloud Infrastructure Foundations Associate, Docker Foundations Professional Certificate

EXPERIENCE

Software Engineer Intern

June 2026 - August 2026

Mastercard

St. Louis, MO

- Developed a real-time monitoring dashboard using **Java**, **Spring Boot**, and **React** to centralize microservice metadata, reducing production deployment cycles by **4+** hours
- Modernized a legacy ingestion engine by prototyping a **Spark**-based distributed parallel processing model, constructed to scale volume for **100GB+** files and **5000+** daily financial reports
- Optimized core report processing engine by pioneering a hybrid **Java-Python** solution, eliminating a **2.5x PySpark** loading overhead and delivering a **40%** efficiency gain across standard operational runs

Undergraduate Teaching Assistant - Building User Interfaces

September 2025 - Present

University of Wisconsin-Madison

Madison, WI

- Mentored CS571 students by resolving complex conceptual and technical challenges in weekly projects using **React** or **React Native**, while also providing guidance on **JavaScript**, **HTML**, **Bootstrap**, and UI/UX principles

Software Engineer Intern

May 2025 – August 2025

State Farm

Bloomington, IL

- Architected an internal tool featuring an embedded AI chatbot with a user-friendly **React** interface, utilizing **Terraform** for scalable infrastructure
- Engineered efficient **Python**-based **REST** APIs within the **AWS** ecosystem, reducing redundant API calls across application by **95%**
- Accelerated workflow by implementing a **CI/CD** pipeline, enforcing code quality by maintaining over **90%** unit testing coverage with **Jest** and **Pytest**

Research Computing Facilitator Assistant

July 2024 – Present

University of Wisconsin-Madison Center of High Throughput Computing

Madison, WI

- Streamlined research computing data by using **Python**, **Pandas**, and **Bash** to automate data collection and report generation, optimizing workflow efficiency by **20%**
- Generated insightful reports and tables from key metrics using **R** and **Matplotlib**, facilitating analysis of high throughput computing usage

PROJECTS

FragranceAI

- Designed a robust data pipeline for **50,000+** fragrance entries, leveraging advanced text feature engineering and efficient batch processing of embeddings to build a high-performance **ChromaDB** vector store for semantic search
- Developed a scalable microservice architecture for AI powered APIs using **FastAPI**, incorporating **Retrieval-Augmented Generation**

LEADERSHIP

Computer Hardware Engineer

May 2024 – August 2024

NASA

Remote

- Spearheaded the design, selection, and integration of the Command and Data Handling (CDH) subsystem of a conceptual lunar rover mission in a student-led team of **10+** students